Curvature-Bounded Black Holes and Compact Objects in CPTG

Strong-Field Admissibility, Polarization–Transport Crossover, and Research Roadmap

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Abstract

Curvature Polarization Transport Gravity (CPTG) treats black holes and other extreme compact systems through the same gravitational hierarchy used elsewhere in the theory: General Relativity (GR), curvature polarization, and curvature transport. This paper does not claim a completed microscopic interior or a unique compact action. It defines the admissible strong-field solution class and the roadmap for deriving it. GR remains the controlling description wherever the additional CPTG response is negligible, including ordinary exterior Kerr or Schwarzschild geometry and the associated horizon structure. As the local geometry becomes increasingly extreme, polarization and transport may grow smoothly from a negligible correction into a leading response before the corresponding classical continuation reaches an inadmissible singular or unbounded-curvature regime. The location and width of this polarization–transport crossover are object dependent and are not assigned a universal radius, curvature condition, or mass scale. CPTG also distinguishes localized compact-source mass from exterior gravitational charge. The bookkeeping difference between them is attributed to the solved geometric response, so a positive geometric contribution permits less localized matter than the exterior GR mass coordinate alone would suggest. No stellar-scale lower source mass is imposed on the admissible black-hole class. Stellar remnants, quiet-collapse and vanishing-star candidates, merger remnants, supermassive compact objects, very low-source-mass compact objects, and primordial compact objects are therefore allowed research classes, subject to conservation, finite curvature, and recovery of the appropriate GR exterior. Specific interiors, crossover laws, mass-conversion factors, primordial abundances, perturbative signatures, and evaporation limits remain future calculations.

1 Introduction

Black-hole physics does not require CPTG to replace General Relativity. GR already provides the successful exterior description of stationary black holes, orbital dynamics, gravitational lensing, horizon formation, and gravitational-wave propagation. CPTG adds a narrower admissibility condition: the physical compact continuation must remain finite. The Einstein geometry remains present throughout; what can change in the deepest strong-field regime is the dynamical importance of curvature polarization and curvature transport. The relevant crossover is therefore between a GR-dominated regime and a response-dominated regime within CPTG, not between two disconnected theories.

This paper is a theory-position and roadmap paper. It does not promote a particular polynomial interior, auxiliary field, matching radius, or critical gradient to parent-theory authority. Its purpose is to state what compact objects CPTG allows, what it forbids, and what future calculations must establish before any specific realization becomes a prediction.

The observational motivation spans several formation channels. GW190814 and GW230529 illustrate compact systems near the traditional neutron-star/black-hole mass boundary [1, 2]. Vanishing-star candidates such as NGC 6946-BH1 and M31-2014-DS1 constrain nonluminous compact-remnant formation through stellar disappearance [3, 4, 5]. Early-universe compact objects provide a separate possible formation channel. CPTG permits all of these as research classes while presently assigning none a unique interior, universal crossover scale, universal source-mass threshold, or universal observational signature.

2 Parent CPTG Strong-Field Architecture

The gravitational field equation remains

$$G_{\mu\nu} + \Pi^{\mathrm{pol}}_{\mu\nu} + \Theta^{\mathrm{trans}}_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}. \tag{1}$$

The ordinary physical stress–energy remains on the right-hand side. Curvature polarization and curvature transport are geometric response terms, not dark-matter fluids and not additional material species.

The relationship to GR is equally direct. Wherever the additional CPTG response becomes negligible,

$$\Pi^{\mathrm{pol}}_{\mu\nu} + \Theta^{\mathrm{trans}}_{\mu\nu} \to 0, \tag{2}$$

Eq. (1) reduces to the Einstein equation with the same physical source. A black-hole horizon is therefore not, by itself, a boundary between GR and CPTG. The exterior and the ordinary horizon can remain completely GR controlled while polarization and transport become important only deeper inside the compact solution.

The strong-field problem is consequently one of continuation, not replacement. CPTG asks how the polarization–transport response grows from negligible to dynamically important before the corresponding classical continuation becomes physically inadmissible. Nothing in the parent architecture requires a sharp universal switch. The natural expectation is an object-dependent crossover in dynamical dominance, whose location and width must ultimately be selected by the compact field equations.

3 Compact-Object Admissibility Rules

The compact sector is governed by a small set of theory rules. These rules define the allowed solution class before any detailed interior model is selected.

3.1 Finite physical solution class

A physical CPTG black hole or compact remnant must remain in a finite-curvature solution class. A singular or unbounded-curvature central continuation is not admitted as the completed physical state. This is an admissibility requirement on CPTG solutions, not a statement that the Einstein equations are discarded or that GR ceases to describe the surrounding spacetime.

3.2 Ordinary horizons remain allowed

CPTG does not remove black-hole horizons. Event and trapping horizons are geometric properties of the solved spacetime and remain part of the allowed compact solution. The theory therefore permits objects that are externally indistinguishable from ordinary Kerr or Schwarzschild black holes while differing only in their deeper physical continuation.

3.3 Exact GR recovery is required where the response becomes negligible

Where the polarization–transport response becomes negligible in vacuum, the solution must return to the corresponding GR vacuum branch. A proposed compact realization that requires a permanent non-GR far-field tail merely to keep its interior finite is not the minimal CPTG black-hole interpretation developed here.

3.4 The strong-field response crossover is local and smooth

The compact response is not assigned to a universal radius such as one gravitational radius, nor is the ordinary horizon declared to be a CPTG activation surface. The theory permits an overlap region in which GR remains an excellent leading description while curvature polarization and transport grow continuously from negligible to dynamically important. This crossover is controlled by the solved local strong-field state rather than by the total mass class alone.

This point matters across scales. A supermassive black hole can possess comparatively mild curvature at its outer horizon and may remain GR dominated far into the trapped region before the compact response becomes important. A much smaller black hole can reach extreme curvature at a much larger fraction of its gravitational size. CPTG therefore allows the polarization–transport crossover to lie relatively deeper in very massive objects and relatively closer to the trapping scale in very small objects. This scale dependence is an organizing expectation, not a presently derived threshold law.

3.5 The physical source mass need not equal the exterior mass

CPTG distinguishes the physical compact-source contribution from the asymptotic gravitational charge. The bookkeeping statement is

$$M_\infty = M_\mathrm{c} + M_\mathrm{geom}, \tag{3}$$

where $M_\mathrm{geom} \equiv M_\infty - M_\mathrm{c}$ is a bookkeeping coordinate for the contribution attributed to the solved geometric response. Equation (3) permits $M_\mathrm{c} < M_\infty$ when that contribution is positive. This statement does not claim a unique local gravitational-energy density or a universal decomposition of the interior; both the covariant localization and the mass ratio remain outputs of future compact solutions.

3.6 No stellar-scale lower source mass is assumed

The admissible black-hole class is defined by the solved trapped geometry and finite strong-field continuation, not by a requirement that the source contain a stellar amount of matter. Very small physical source masses may therefore belong to the CPTG black-hole class if their geometry reaches a trapped strong-field state and the full CPTG response supplies a consistent finite continuation. Whether semiclassical physics or the completed compact equations impose a lower mass in some regime is a separate future question; no such lower bound is assumed in this paper.

3.7 No universal compact interior is assumed

Different compact objects may realize different finite internal structures while sharing the same parent CPTG architecture and the same exterior GR requirements. The theory therefore does not require every stellar black hole, supermassive black hole, merger remnant, or primordial compact object to possess one common density profile, one common matching radius, or one common polarization fraction.

3.8 Formation history and stationary structure are distinct problems

The matter and radiation involved in collapse determine the formation history, emitted energy, fallback, angular momentum, and conserved charges. The stationary compact solution is the gravitational state selected after that dynamical history. A complete future calculation must connect the two, but the present admissibility framework does not require the stationary object to be modeled as an ordinary hydrostatic star hidden behind a horizon.

4 Finite Central Continuation in CPTG

A CPTG black hole is an ordinary black hole in its GR-dominated domain together with a finite deeper continuation. The phrase "finite core" is used only as shorthand for this requirement; it does not commit the theory to a material ball, a de Sitter core, a gravastar-like state, or any other specific composition. The central physical continuation must remain finite in curvature, while a smooth polarization–transport crossover may occupy the region between the GR-dominated geometry and the deepest response-dominated state.

For a nonrotating object the exterior may remain exactly Schwarzschild wherever the additional CPTG response is negligible; for a rotating object the corresponding exterior may remain exactly Kerr. The ordinary outer horizon is retained. The response onset need not coincide with that horizon and need not occur at the same fractional radius for objects of different mass. Any inner trapping structure, crossover region, or central geometry must be derived from the completed compact field equations rather than assigned from a preferred analytic interpolation.

Mathematically regular interior metrics remain useful exploratory examples, but their existence establishes possibility rather than CPTG selection. The authority of this paper is therefore the finite solution class and the conditions future compact solutions must satisfy.

5 Mass Interpretation and the Lower Mass Gap

The mass measured from an exterior binary orbit, lensing field, or gravitational-wave signal is an exterior gravitational charge. CPTG permits that charge to differ from the localized compact-source mass because the solved polarization–transport geometry also contributes to the gravitational response. Consequently, an object with exterior mass M_∞ need not contain the same amount of localized physical source mass. The larger exterior field is not an observational illusion; what remains to be derived is the covariant compact solution that fixes the difference $M_\infty - M_c$.

This possibility is especially relevant to lower-mass-gap systems. GW190814 contained a secondary object reported near $2.6M_\odot$, and GW230529 further emphasized the astrophysical importance of compact objects near the traditional neutron-star/black-hole boundary [1, 2]. CPTG does not currently predict that either object must be a finite-core black hole, nor does this paper assign either one a reduced physical source mass. They are observational targets for the future mass-conversion problem.

The required future calculation must keep separate the progenitor mass, terminal stellar mass, ejected mass, fallback, radiative and neutrino losses, physical post-collapse compact-source mass, and exterior gravitational charge. A CPTG mass conversion becomes a prediction only when those quantities are connected by the completed compact formation and field equations. Until then, the strong claim is only that less physical source mass is allowed when a positive geometric contribution supplies part of the exterior gravitational charge. The possibility is not restricted to stellar-mass systems; it is potentially most consequential for compact objects formed from comparatively small amounts of matter, where the geometric response could constitute a large share of the exterior charge.

6 Allowed Compact Formation Channels

CPTG does not restrict finite compact objects to one astrophysical origin. Each admissible realization must satisfy the same finite-curvature, conservation, and GR-recovery requirements.

6.1 Quiet stellar collapse and vanishing stars

NGC 6946-BH1 remains a major failed-supernova or vanishing-star candidate, while M31-2014-DS1 provides another massive-star disappearance case associated with black-hole formation [3, 4, 5]. These observations do not measure a CPTG interior; they are formation candidates for future analyses that keep progenitor, ejecta/fallback, compact-source, and geometric-response contributions distinct.

6.2 Ordinary stellar and merger formation

Black holes formed through core collapse, fallback, binary interaction, or compact-object mergers are admissible provided the final geometry satisfies the finite-solution condition. Their stationary structure may depend on mass, angular momentum, matter content, transient transport, and radiative losses, but this paper assigns no universal post-collapse or post-merger core and no prompt-collapse threshold.

6.3 Low-physical-mass and nonstellar compact objects

CPTG also permits black-hole-like compact states whose localized source mass is far below ordinary stellar-remnant scales. The admissibility question is geometric: whether the source and boundary data produce a trapped solution with a finite polarization–transport continuation and the appropriate GR exterior. Their formation, stability, evaporation behavior, and any lower mass boundary remain future compact and semiclassical calculations.

6.4 Supermassive compact objects

The same rules extend to supermassive objects. Because the local curvature scale associated with a very massive horizon can be comparatively mild, the GR-dominated region may extend relatively deep before polarization and transport become important. A finite central continuation, ordinary GR exterior, and object-dependent response crossover remain the organizing principles.

6.5 Primordial compact objects

CPTG permits primordial compact objects formed before ordinary stellar evolution, including possible very low-source-mass states, as an allowed future compact population [6, 7]. Their formation mechanism, mass function, survival, merger history, and abundance must be derived within the native CPTG cosmological framework rather than imported from a dark-matter or FLRW-based prescription. No primordial abundance or dark-matter fraction is claimed here; the present statement is only one of admissibility.

7 Current Compact-Sector Status

The theory position can be summarized without assigning unsupported numerical closures.

Table 1: CPTG compact-sector admissibility and current claim boundaries.

Topic	**Current CPTG position**	**What remains future work**
Finite central continuation	Required admissible physical solution class	Derive the selected finite interior from the compact field equations
GR exterior and outer horizon	Retained wherever polarization and transport are negligible	Verify rotating and dynamical recovery of the GR branch
Strong-field response crossover	Expected to be smooth, local, and object dependent rather than fixed at a universal radius	Derive how polarization and transport grow from negligible to dynamically important
Physical source mass below exterior mass	Allowed when geometric response contributes positively; the exterior charge need not be entirely material	Derive the mass conversion from source state, spin, and compact dynamics
Vanishing-star remnants	Allowed CPTG formation candidates	Perform object-specific source, fallback, and remnant calculations
Lower-mass-gap compact objects	Observational targets	Determine whether any require or favor a CPTG mass conversion
Low-physical-mass compact objects	Allowed; no stellar-scale minimum source mass is imposed by this roadmap	Determine formation, stability, semiclassical limits, and the geometric share of the exterior charge
Merger remnants	Allowed finite compact states	Solve dynamical polarization/transport during and after merger
Supermassive compact objects	Allowed under the same finite-solution rules; the crossover may lie relatively deep	Derive scale-dependent compact response and growth history
Primordial compact objects	Allowed early-universe compact population	Derive formation, mass function, survival, abundance, and lensing within native CPTG cosmology

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Topic	Current CPTG position	What remains future work
Universal interior profile	Not claimed	Determine whether the full theory selects one family or multiple finite families
Universal core or crossover radius	Not claimed	Derive object-dependent crossover and finite central scales from the field equations
Universal crossover condition	Not claimed	Determine the local geometric state that controls the compact response without presupposing a fixed threshold
Universal mass-amplification law	Not claimed	Derive $M_c \leftrightarrow M_\infty$ from the compact solution
Compact covariant response law	Not yet derived	Derive the strong-field polarization and transport tensors from parent CPTG
Echoes, tidal response, or modified ringdown	Not automatic consequences	Compute perturbations only after the background compact solution is derived
Evaporation endpoint or remnant mass	Not claimed	Requires a separate CPTG semiclassical/thermodynamic treatment

8 Observational Program

Formation tests ask how much localized source mass and angular momentum are required to produce a compact object of a given exterior mass and spin. Vanishing stars, failed-supernova candidates, lower-mass-gap binaries, and merger remnants are useful because they constrain source history rather than only the final exterior field. Strong-field tests instead ask whether a finite CPTG continuation leaves departures from the corresponding GR exterior. Such departures are not guaranteed: a very massive object may hide the polarization–transport crossover deep inside the trapped region, while in a sufficiently small object the same crossover may occupy a larger fraction of the compact geometry. Ringdown, tidal response, secondary pulses, and merger differences therefore remain downstream calculations on a solved background.

Population tests apply primarily to primordial and other nonstellar compact objects. They require CPTG-native formation histories, mass functions, merger evolution, and observational selection before comparison with microlensing, gravitational-wave, accretion, and structure constraints. No such population calculation is assumed by this roadmap.

9 Claims Deliberately Deferred

This paper does not adopt a specific analytic interior, fixed matching radius, universal inner-horizon location, compact critical gradient, universal crossover condition, universal compact acceleration normalization, auxiliary state field, mass-amplification law, or lower black-hole source mass as parent-theory authority. It likewise makes no present prediction for tidal deformability, gravitational-wave echoes, enhanced ejecta, information storage, an evaporation freezeout mass, or primordial compact-object abundance.

These are deliberately separated from the admissibility rules. Future derivations may recover some explored candidate structures or select different finite continuations. The roadmap is intended to remain valid either way.

10 Research Roadmap

The compact program has two foundational mathematical tasks. They should be completed before detailed phenomenology is treated as theory authority.

The first priority is to derive the compact strong-field forms of the parent polarization and transport tensors in a covariant manner that remains regular through trapped regions. The goal is a closed field system showing how the additional response grows smoothly from a GR-dominated regime, produces a finite continuation before the corresponding classical branch becomes physically inadmissible, and returns to exact GR wherever the response becomes negligible.

The second priority is the stationary spherical boundary-value problem across a broad mass range. That calculation must determine whether CPTG selects one finite interior family or several, where and how broadly the polarization–transport crossover develops for very small and very massive objects, how the exterior gravitational charge divides between localized source mass and geometric response, and what trapping structure follows from the selected solution.

Once those two foundations are closed, the downstream program is ordered rather than independent. Rotation should lift the spherical response to stationary axisymmetry and recover the Kerr exterior with mass, angular momentum, horizon structure, and crossover scales emerging from one solution. Dynamical formation should then connect stellar collapse, fallback, mergers, primordial collapse, and low-source-mass trapping to the stationary compact state. Perturbation theory can subsequently determine ringdown, tidal response, or any secondary wave structure. Population calculations for primordial and other nonstellar compact objects come last, using CPTG-native formation and abundance histories before comparison with microlensing, gravitational-wave, accretion, and structure constraints.

11 Failure Conditions

The strong-field program is falsifiable even before every observable is calculated. A completed CPTG compact solution must remain finite, respect the parent conservation structure, possess a well-defined and smooth polarization–transport response through any trapped region, and recover the appropriate GR exterior where that response becomes negligible. If the completed theory cannot produce a globally consistent finite solution for physically relevant black-hole charges, or if its physical continuation necessarily enters an unbounded-curvature regime, the compact-sector CPTG position fails in that domain.

A proposed mass conversion also fails if it cannot be derived from the compact source and geometric response while preserving conservation and the measured exterior charges. Likewise, a proposed primordial compact population fails if its CPTG-native formation and abundance calculation conflicts with the observational limits relevant to that population.

These are theory-level failure conditions. They do not require this roadmap paper to solve every future boundary-value, collapse, perturbation, or population problem in advance.

12 Conclusion

The CPTG strong-field position does not require a second theory of black holes. GR controls the exterior spacetime and ordinary horizon physics wherever the additional geometric response is negligible. In the deeper compact regime, curvature polarization and transport may become progressively important through a smooth, object-dependent crossover and must select a finite physical continuation. The crossover is not assigned to a universal radius or threshold and need not occupy the same fraction of the geometry in a very small black hole and a supermassive one.

This solution class permits stellar and merger remnants, quiet-collapse and vanishing-star candidates, supermassive objects, very low-source-mass compact objects, and primordial compact objects. It also permits localized source mass to be smaller than the exterior gravitational charge when the solved geometric response contributes positively. The governing roadmap is therefore clear: derive the compact covariant response first, solve the spherical boundary-value problem across mass scales second, and treat rotation, formation, perturbations, and populations as downstream calculations. No particular interior profile, amplification factor, minimum source mass, or primordial abundance is required for the present theory position.

Data, Code, and Evidence Availability

This paper is a theory-position and research-roadmap manuscript. Its compact-sector claims are qualitative admissibility statements derived from the parent CPTG architecture together with cited observational examples. No fitted compact radius, numerical interior profile, universal crossover condition, universal lower source mass, universal mass-amplification factor, primordial abundance function, echo amplitude, tidal deformability, or evaporation endpoint is used as a theory input or claimed as a result.

The parent CPTG framework and supporting public research materials are maintained through the project repository [8]. The vanishing-star and gravitational-wave examples used as observational motivation are drawn from the cited literature. Future quantitative compact-sector papers should provide their own numerical evidence records for the specific boundary-value, formation, perturbation, or population calculations they perform.

References

[1] R. Abbott et al. [LIGO Scientific Collaboration and Virgo Collaboration], *GW190814: Gravitational Waves from the Coalescence of a 23 Solar Mass Black Hole with a 2.6 Solar Mass Compact Object*, Astrophys. J. Lett. **896**, L44 (2020). DOI: 10.3847/2041-8213/ab960f.

[2] LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, *GW230529: Observation of the Merger of a Neutron Star and an Unknown Compact Object*, public LVK release (2024). Official LVK science summary.

[3] S. M. Adams, C. S. Kochanek, J. R. Gerke, K. Z. Stanek, and X. Dai, *The Search for Failed Supernovae with the Large Binocular Telescope: Confirmation of a Disappearing Star*, Mon. Not. R. Astron. Soc. **468**, 4968–4981 (2017). DOI: 10.1093/mnras/stx816.

[4] C. M. Basinger, C. S. Kochanek, S. M. Adams, X. Dai, and K. Z. Stanek, *The Search for Failed Supernovae with the Large Binocular Telescope: N6946-BH1, Still No Star*, Mon. Not. R. Astron. Soc. **508**, 1156–1164 (2021). DOI: 10.1093/mnras/stab2620.

[5] K. De et al., *Disappearance of a Massive Star in the Andromeda Galaxy Due to Formation of a Black Hole*, Science **391**, 689–693 (2026). DOI: 10.1126/science.adt4853.

[6] S. W. Hawking, *Gravitationally Collapsed Objects of Very Low Mass*, Mon. Not. R. Astron. Soc. **152**, 75–78 (1971). DOI: 10.1093/mnras/152.1.75.

[7] B. J. Carr and S. W. Hawking, *Black Holes in the Early Universe*, Mon. Not. R. Astron. Soc. **168**, 399–415 (1974). DOI: 10.1093/mnras/168.2.399.

[8] Carter L. Glass Jr., *Curvature Polarization Transport Gravity: A Unified Geometric Framework for Structure, Cosmology, and Nuclear Reactions*, 2026, DOI: 10.5281/zenodo.22441039.

[9] CPTG, Supporting Python Models, Benchmark Implementations, and Research References for Curvature Polarization Transport Gravity, companion resource for the present work, available at `https://github.com/CLG2025/CPTG`.